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Charging and its mitigation in the proof masses of LISA Pathfinder

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Abstract

The precise measurement of gravitational waves in missions like LISA Pathfinder is limited by spurious forces acting on the free falling test masses. A primary source of this disturbance is electrostatic charging caused by the constant bombardment of high energy galactic cosmic rays and solar energetic particles. This review examines the physical mechanisms behind test mass charging and its consequences. A catalogue of the principal mitigation solutions developed to counteract this effect is compiled, spanning both active and passive techniques. Effective charge management is a critical technology for enabling the science goals of current and future space based gravitational wave observatories.

Keywords LISA, Test mass, gravitational waves, charging, interferometer, detectors

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2 Introduction

The Laser Interferometer Space Antenna (LISA) mission represents an immense effort to observe the universe in a new light by detecting low frequency gravitational waves. Unlike ground based detectors which are limited by seismic noise a space based observatory can access a largely unexplored part of the gravitational spectrum. The success of LISA and its precursor mission LISA Pathfinder (LISA PF) ([Figure 1](#)) depends on an incredibly precise measurement technique: tracking the motion of free falling objects known as test masses (TMs) as they travel through space.

These TMs are designed to follow perfect geodesic paths influenced only by gravity. The spacecraft is carefully built around them, shielding them from external forces and using sensitive instruments to monitor their position. The LISA PF employs these TMs as end-mirrors in a highly sensitive interferometric setup, where the relative displacement between the masses is monitored with extreme precision that will detect the strain induced by gravitational waves. However, the space environment itself presents a major challenge. The constant bombardment by high energy particles from the sun and deep space causes electrical charge to accumulate on the surfaces of these isolated proof masses.

This buildup of charge which is a process known as test mass charging, introduces spurious electrostatic forces that can push and pull on the masses. These forces disturb their perfect free fall, creating acceleration noise that can easily engulf the faint signals from gravitational waves. Therefore, the sensitivity required by the mission is directly affected by this charging effect. Understanding and controlling this phenomenon is not just a technical detail, it is an obstacle that must be overcome for the mission to succeed.

This literature review will explore the research dedicated to this central problem. The primary goal is to understand the physical mechanisms behind the charging of the proof masses and to investigate the mitigation strategies that have been developed to counteract it. By examining the work on this topic this review aims to provide a foundational understanding of the challenges and catalogue of solutions related to ensuring the TMs remain in a state of near perfect free fall.



Figure 1: LISA Pathfinder (2 meters tall and about 1.2 meters in diameter) [1]

3 Scientific motivation

3.1 Gravitational wave detection

Gravitational wave detection represents a transformative advance in our understanding of the universe, offering a fundamentally new observational window that complements traditional electromagnetic astronomy. The direct measurement of gravitational waves enables the study of phenomena that are otherwise inaccessible, such as the coalescence of black holes and other cataclysmic astrophysical events. These waves, predicted by Einstein's general theory of relativity, carry information about their sources and the nature of gravity itself, propagating virtually unimpeded through matter and thus providing insights into regions of the cosmos that are opaque to electromagnetic radiation.

The Laser Interferometer Space Antenna (LISA) and its precursor, LISA Pathfinder, are designed to detect gravitational waves in the low-frequency regime, specifically in the range of 10^{-4} to 0.1 Hz as this frequency band is particularly rich in signals from compact stellar binaries, extreme mass-ratio black-hole binaries, massive black-hole mergers (Figure 2i) and possibly even stochastic backgrounds from the early universe. The scientific motivation for gravitational wave detection goes beyond confirmation of just general relativity. It provides a direct probe of strong-field gravity, tests the propagation speed of gravitational waves and allows for the measurement of cosmological parameters that are independent of electromagnetic observations [2]. Furthermore, gravitational wave astronomy has the potential to uncover entirely new classes of astrophysical objects and phenomena as it is sensitive to sources that may be electromagnetically dark.



Figure 2: i) Two black holes merge into one, ii) Laser Interferometer Gravitational-wave Observatory (LIGO) [3]

3.2 Challenges in terrestrial detection

The detection of gravitational waves on Earth is fundamentally constrained by a variety of environmental and instrumental noise sources which collectively limit the sensitivity of terrestrial observatories especially at low frequencies that become inaccessible. Ground based detectors such as LIGO (Figure 2ii) are subject to seismic noise, thermal fluctuations and gravity gradient noise, all of which are challenging to suppress below the levels required for the detection of weak gravitational wave signals.

Seismic noise arises from natural and anthropogenic ground vibrations. This residual motion from the Earth's surface couples into the detector, even with advanced isolation systems, setting a lower bound on the accessible frequency range. Gravity gradient noise, also known as Newtonian noise, arises from fluctuating gravitational fields produced by moving masses near the detector such as atmospheric density variations, groundwater flow or human activity. Unlike seismic noise, gravity gradient noise cannot be shielded by mechanical isolation as it is a direct gravitational effect. This sets a fundamental limit to the sensitivity of ground-based detectors at low frequencies. Thermal noise originating from the Brownian motion of atoms in the detector components, further restricts the sensitivity, particularly in the suspension systems and mirror coatings. These ground based detectors also have to deal with other modes of noise such as residual gas noise and quantum noise.

While cryogenic cooling, smarter architecture and material engineering can mitigate some of these effects, they cannot be eliminated entirely and their impact becomes more pronounced as detectors strive for higher sensitivity. The cumulative effect of these noise sources is that ground based detectors are effectively blind to gravitational waves below a few Hertz. The measurement bandwidth of future space-based observatories such as LISA is specifically chosen to target the frequency range inaccessible to ground-based detectors due to these noise sources [3].

3.3 Need for space based observatory

LISA and its precursor, LISA PF, are designed to overcome these terrestrial limitations by operating in the quiet environment of space where seismic and gravity gradient noise are absent and thermal fluctuations can be more effectively controlled. While the transition to space based detection introduces challenges like the charging of TMs, it provides access to the ultra quiet conditions of the millihertz frequency window which is fundamentally inaccessible from the ground. This environment enables the high sensitivity required to detect low frequency gravitational waves, opening new avenues for astrophysical discovery. By placing the observatory in space, it becomes possible to isolate and allow the detection of minute spacetime distortions caused by passing gravitational waves.

While this introduces new challenges, it is still a manageable trade-off for the significant reduction in environmental noise compared to ground-based observatories. Furthermore, the ability to deploy long interferometric arms spanning millions of kilometers in LISA is only possible in space. This extended baseline is essential for achieving the sensitivity required to detect low frequency gravitational waves. On Earth, such large scale infrastructure is not feasible due to geographical and engineering constraints [4].

3.4 Role of LISA Pathfinder

LISA PF was conceived as a dedicated technology demonstrator to validate the core experimental techniques and hardware required for LISA, which will be the first space-based gravitational wave observatory. A central objective of LISA PF was to demonstrate the ability to maintain TMs in a near-perfect free-fall state, isolated from all non-gravitational forces to an unprecedented degree. This is essential for LISA, as the detection of gravitational waves relies on measuring minute changes in the separation between free-falling TMs housed in separate spacecraft using laser interferometry. The LISA PF mission architecture included two gold-platinum alloy cubes serving as TMs, housed within inertial sensors, were designed to be shielded from external forces, with the spacecraft itself actively maneuvering to follow the motion of one of the masses.

One of the most significant sources of disturbance in the space environment is the charging of the TMs. LISA PF provided a unique platform to study these charging mechanisms in situ, allowing for the validation of theoretical models and simulations that predict the rates and effects of test mass charging under realistic space conditions. The mission also enabled the development and testing of active charge management systems which are essential for maintaining the TMs at a controlled potential. Furthermore, LISA PF served as a testbed for the integration and performance assessment of the subsystems required for LISA, including the laser interferometry system, the inertial sensors and the drag-free control algorithms. The successful operation of LISA PF thus provided empirical evidence that the technological and methodological foundations for LISA are sound and that the mission concept is feasible within the constraints imposed by the space environment.

4 LISA Pathfinder Overview

4.1 Mission Objectives

The primary objective of LISA PF was to achieve and characterize the extremely low levels of acceleration noise necessary for the detection of gravitational waves in space. By isolating free-falling TMs from all non-gravitational forces, LISA PF aimed to replicate the drag-free conditions essential for LISA's scientific goals. The TMs served as proxies for the mirrors in LISA, whose relative positions must be measured with picometer precision to detect the minute spacetime distortions caused by passing gravitational waves. The spacecraft was equipped with a sub-picometer laser interferometer to monitor the differential motion of the TMs (38 cm apart), thereby quantifying the residual acceleration noise and identifying its sources [5].

A significant achievement of LISA PF was the demonstration that the acceleration noise acting on the TMs could be reduced to levels compatible with, or even surpassing the requirements for LISA in the frequency band between 0.1 mHz and 0.1 Hz. This was accomplished through a combination of advanced charge management techniques and careful control of the spacecraft environment. The LISA PF mission yielded valuable insights into the behavior of low-energy electromagnetic processes and their impact on TM charging. The observed charging rates and their correlation with solar activity proxies (solar flares and solar energetic particles) provided a robust empirical basis for refining the models used to forecast the space environment's influence on LISA. The mission's data enabled the identification of both steady-state and transient charging events, including those associated with solar energetic particle bursts and informed the development of operational strategies to maintain optimal test mass neutrality throughout the mission duration.

The lessons learned from LISA PF regarding spacecraft shielding, charge mitigation and environmental monitoring have been directly incorporated into the LISA mission architecture. The empirical data collected by LISA PF have also contributed to a deeper understanding of the interplay between cosmic radiation, solar activity and test mass charging, informing both mission planning and real time operational protocols for future gravitational wave observatories [6].

4.2 Gravitational Reference Sensors

At the heart of this inertial sensor design are the free-falling TMs, which are housed within the spacecraft and are surrounded by electrode housings (18 gold electrodes) that serve both as capacitive sensors and actuators, enabling precise measurement and control of the TM positions ([Figure 3](#)). The electrodes are arranged to allow for the detection of minute displacements in all translational and rotational degrees of freedom, which is essential for maintaining the TM in a near perfect free fall state. The TM was entirely detached from its environment with no electrical connection to the ground [7]. The gold coatings on both the TMs and are processed to achieve work functions in the range of 3.9–4.5 eV which is critical for controlling photoelectric emission and minimizing unwanted charging. These surfaces are baked to 115°C in vacuum and stored under controlled conditions prior to launch to ensure their stability, reliability and cleanliness [8]. Furthermore, to mitigate unwanted edge effects there is a an equipotential protection ring at the periphery [9].

4.3 Environment in space

The spacecraft was initially deployed by Vega flight VV06 into an elliptical low Earth orbit. Subsequently, it performed brief burns at each perigee, incrementally elevating the apogee toward the desired halo orbit at the Earth–Sun Lagrange point L1 [1]. In space the spacecraft is exposed to a continuous flux of high-energy particles originating from galactic cosmic rays (GCRs) and solar energetic particles (SEPs). These particles can penetrate the spacecraft's shielding and interact

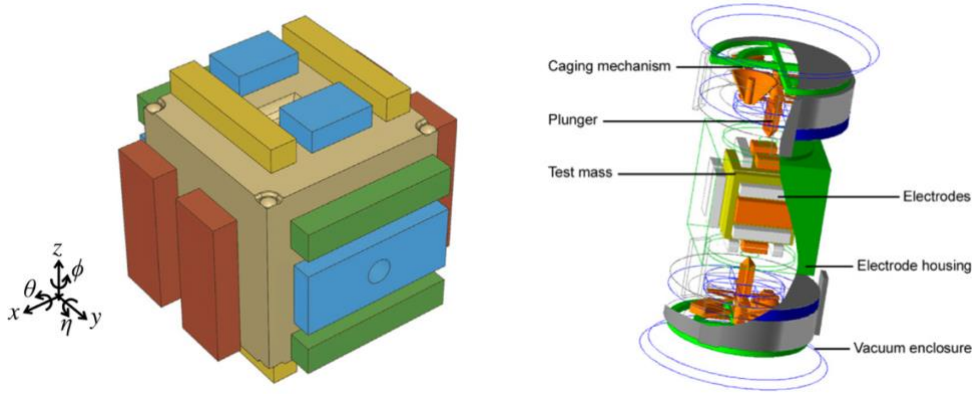


Figure 3: i) Diagram of LISA Pathfinder GRS showing 46 mm cubic test mass surrounded by capacitive-sensing bias injection electrodes in blue; sensing and actuation electrodes in red, green & yellow; ii) caging mechanisms, used to constrain the test masses during launch inside a vacuum enclosure [10]

with the materials inside, particularly the free-floating TMs, leading to the accumulation of electrical charge on their surfaces. The charging of the TMs is a direct consequence of the space environment as cosmic-ray particles and solar protons traverse the spacecraft and deposit charge in the TMs, which can significantly affect the sensitivity of gravitational wave measurements. The energy spectrum of the incident particles is broad with GCRs and SEPs spanning from tens of MeV/nucleon up to several GeV/nucleon. The spacecraft's shielding is effective only up to a certain threshold; for instance, particles with energies below approximately 100 MeV/nucleon are typically stopped within the spacecraft structure and do not reach the TMs while higher-energy particles can penetrate and contribute to charging [11].

The flux and energy distribution of these particles are modulated by solar activity with solar minimum and maximum conditions leading to significant variations in the incident spectra highlight that the solar modulation parameter ϕ is a key factor in determining the GCR flux with lower values during solar minimum resulting in higher GCR intensities and consequently, increased charging rates [12]. In addition to the direct deposition of charge by primary cosmic rays and solar particles, secondary processes such as the emission of low-energy electrons from the surfaces of the TMs and surrounding electrodes further complicate the charging dynamics. The kinetic emission of electrons, triggered by the bombardment of the sensor by energetic particles, introduces additional uncertainties in the net charging rate and can contribute to acceleration noise even if the overall charge is balanced [6].

4.4 Materials used

Mechanical configuration in LISA PF prioritizes the stability and isolation of the TM, as even minute mechanical or thermal fluctuations can introduce noise that compromises measurement precision. The electrode housing and caging mechanism are engineered with high precision and are located in close proximity to the TM as these elements have the most significant influence on the charging and mechanical stability of the proof mass as any mechanical imperfections or misalignments could lead to unwanted forces or torques on the TM. Moreover, the TM and its housing must be maintained at a stable temperature to prevent thermal gradients which can induce mechanical stresses and lead to spurious forces on the TM. The local thermal environment is determined not only by the spacecraft's exposure to solar radiation but also by the heat generated by onboard electronics and the thermal properties of the materials used in the construction of the housing and surrounding structures. Therefore, the choice of materials for the mechanical structure is informed by the need to minimize outgassing, magnetic susceptibility and thermal expansion.

The TMs themselves are typically fabricated from high-purity non-magnetic materials to reduce susceptibility to external electromagnetic disturbances and to limit the accumulation of charge from cosmic rays and solar energetic particles. The surrounding electrode housing and other critical components are constructed from materials with well characterized electrical and mechanical

properties, ensuring that their interaction with incident radiation and secondary particle production is predictable. The distribution and thickness of materials such as the copper shielding used in the radiation monitor are carefully selected to balance the need for protection with the imperative to avoid excessive mass. For example, the radiation monitor employs silicon wafers of 300 μm thickness arranged telescopically and surrounded by a 6.4 mm thick copper box which serves to attenuate the flux of charged particles reaching the sensitive electronics and the TMs [13].

5 Proof Masses in LISA Pathfinder

5.1 Physical properties and Composition

The TMs employed in the LISA PF are designed with requirements such that their physical properties and composition must ensure the highest possible fidelity in gravitational wave detection. The geometry of the TMs is typically cubic which allows for symmetric response to external forces and simplifies the analysis of their motion within the spacecraft. These TMs are fabricated from a high-purity gold-platinum alloy. This alloy is selected for its high density which reduces the influence of residual gas and radiation pressure and for its low magnetic susceptibility which minimizes the impact of ambient magnetic fields. The Au-Pt composition also ensures chemical inertness, reducing the risk of surface contamination or unwanted chemical reactions that could alter the mass or charge state of the TMs.

The TMs are coated with a thin layer of gold, which provides a uniform, conductive and chemically stable surface. This gold coating is essential for maintaining electrical neutrality and for ensuring consistent interaction with the surrounding electrode housing. The lattice spacing at the gold surface is approximately 4.078 Å, a value that is significant when considering the interaction of the TM with incident electrons [8]. For instance, the de Broglie wavelength of an electron at about 9.1 eV matches this lattice spacing, leading to diffraction effects that can influence secondary electron emission and consequently the charging behavior of the TM (Figure 4) [14]. Moreover, the sensing gap along the sensitive axis is designed to be 4 mm, a value chosen to balance the need for high sensitivity in position measurement with the requirement to minimize electrostatic stiffness and associated disturbances. The total capacitance between the TMs and the electrodes, as well as the partial derivatives of capacitance with respect to displacement are determined by this geometry and play a direct role in the electrostatic forces acting on the TM. This is validated through finite element analysis of electrostatic coupling in LISA PF inertial sensors [9].

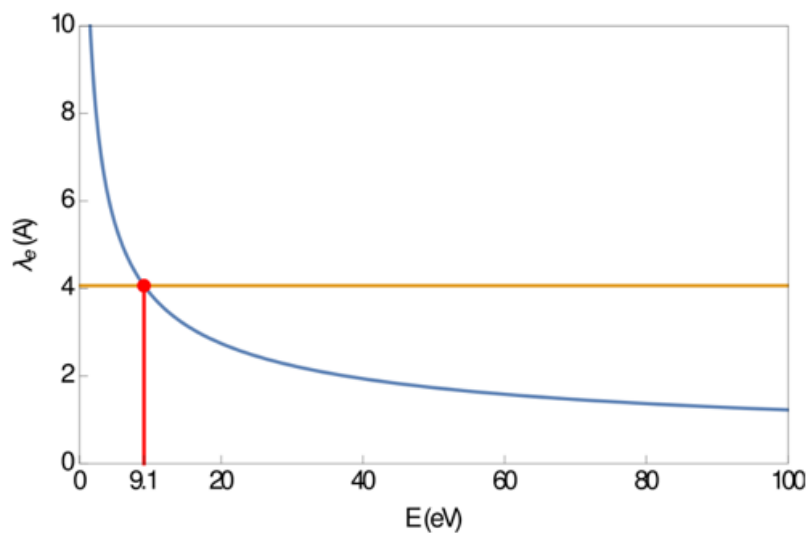


Figure 4: De Broglie wavelength λ_e of electron as a function of electron energy (blue line). The brown line represents the spacing of the Au-Pt lattice. The red line is the energy at which de Broglie wavelength and the lattice spacing are equal $\approx 9.1 \text{ eV}$ [14]

5.2 Suspension and Isolation

The TMs are only in pseudo free fall, as they are subject to disturbances from their immediate environment, which includes the satellite's own fields and thermal gradients. To mitigate these effects, the suspension system must minimize mechanical contact and actively control the position and orientation of the TMs using non contacting methods ([Figure 5i](#)). Electrostatic suspension is a key technique employed in LISA Pathfinder [8]. The TMs are surrounded by electrodes that can apply controlled forces and torques via electrostatic fields, allowing for precise positioning without physical contact. Mechanical isolation is further enhanced by the spacecraft's structure which is engineered to minimize the transmission of vibrations and other mechanical disturbances to the TMs. The use of multiple layers and careful material selection ensures that secondary particles and electromagnetic noise are attenuated before reaching the sensitive components.

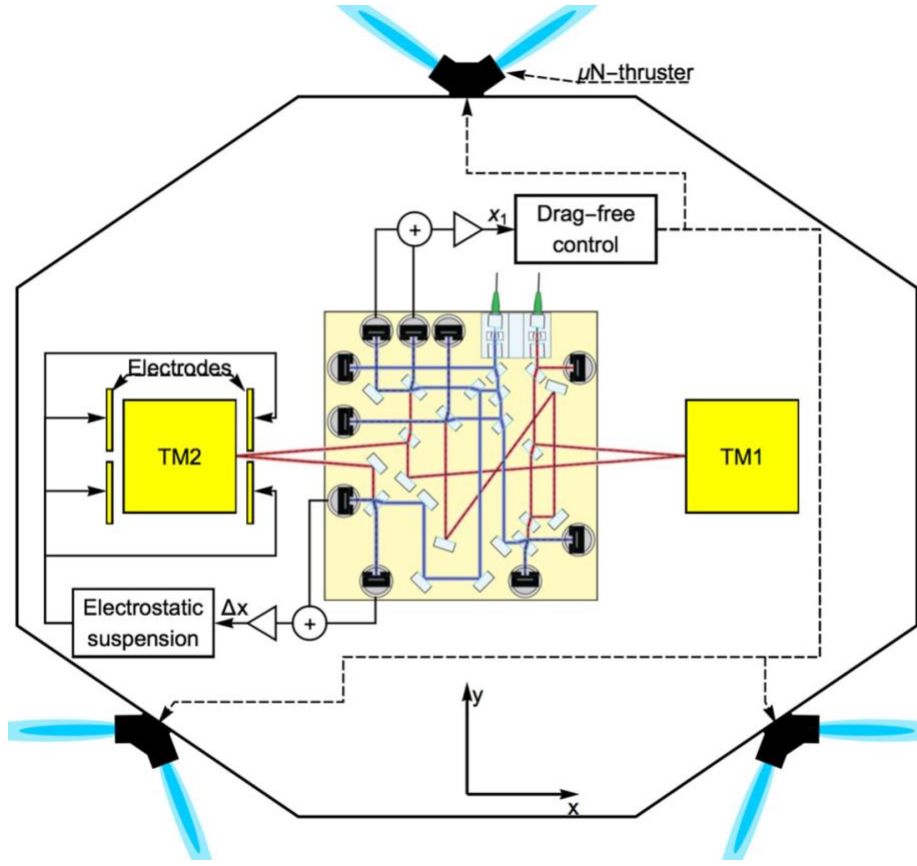
Furthermore, the interplay between photoemission and surface reflectivity, where it is noted that highly reflective surfaces emit few photoelectrons and can thus accumulate significant negative charge in sunlight. This phenomenon necessitates careful placement and design of mirrors and other optical components to prevent unwanted charging and its associated disturbances. Positioning mirrors at a sufficient distance from the satellite surfaces is suggested as a strategy to reduce the impact of their charging on the overall system [15].

5.3 Measurement and Performance Requirements

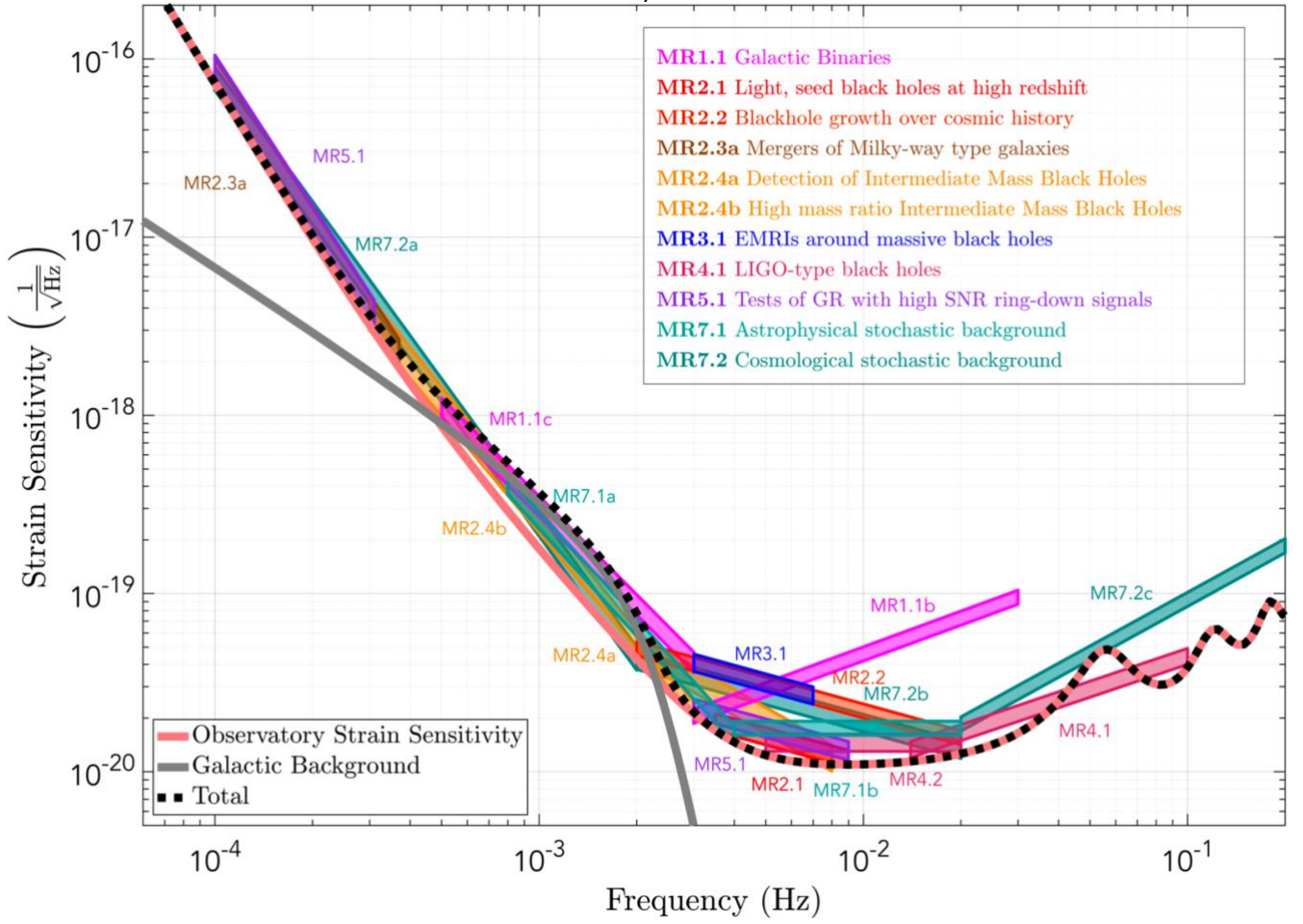
A successful detection requires the signal from a given source to be stronger than the instrument's noise floor. For example, signals from Mergers of Milky-way type galaxies (MR2.3a) or Tests of GR with high SNR ring-down signals (MR5.1) are clearly detectable ([Figure 5ii](#)) at certain frequencies because their characteristic lines lie above the observatory's sensitivity line (the black dotted line). In contrast, other sources like LIGO-type black holes (MR4.1) would be undetectable, as their signal lies almost entirely below this noise floor.

The mission aims to demonstrate that the acceleration noise acting on the TMs can be kept below $3 \times 10^{-15} \text{ m s}^{-2} \text{ Hz}^{-\frac{1}{2}}$ within the frequency band of 10^{-4} to 10^{-1} Hz. This level of precision is necessary to ensure that the instrument's sensitivity is not limited by local disturbances but rather by the fundamental limits imposed by gravitational wave signals themselves. The charging of TMs, primarily due to the interaction with cosmic ray particles and solar energetic particles is recognized as one of the most severe sources of disturbance. These high energy particles can penetrate the spacecraft and deposit charge in the TM bodies [11]. The minimum energy threshold for ions capable of penetrating and charging the TMs is approximately 100 MeV/n, although a conservative threshold of 75 MeV/n was adopted in LISA PF to avoid underestimating the total particle flux and to account for the shielding effect of the spacecraft materials [16]. The charging process is not only a source of direct force noise but also introduces stochastic fluctuations which can manifest as acceleration noise within the measurement bandwidth [6].

Space-based experiments such as LISA PF are essential for accurately characterizing these external space effects as ground-based measurements cannot replicate the cosmic-ray environment encountered in space. Ground-based neutron and muon measurements are insufficient for determining the energy differential fluxes of galactic cosmic rays at the top of the atmosphere whereas space-based magnetic spectrometers provide the necessary data for precise modeling and simulation of the charging processes [13].



i)



ii)

Figure 5: i) Schematic of LISA PF [5] ii) The plot displays the expected signal strength of various astrophysical sources (colored lines) against the total noise sensitivity of the observatory (thick black line). For a successful detection to occur, the signal from a cosmic event must be "louder" than the noise floor; therefore, the signal's line must lie above the observatory's sensitivity line [4]

6 Charging

6.1 Sources of charging

6.1.1 Galactic Cosmic Rays (GCRs)

Galactic cosmic rays (GCRs) are a primary source of charging for the proof masses in LISA PF, significantly influencing the overall charge accumulation and associated noise. GCRs consist mainly of high energy protons and heavier nuclei such as ^4He originating from outside the solar system and propagating through the interstellar medium before reaching the spacecraft. Their flux and energy spectra are modulated by solar activity, for instance, $^3\text{He}/^4\text{He}$ ratio at 1 GeV/nucleon is observed to be approximately 0.22 at solar minimum, dropping to 0.16–0.18 at solar maximum, reflecting the dynamic nature of GCR composition and flux near Earth [17]. When GCRs interact with the gold coated TMs, they induce a variety of physical processes that result in the emission of secondary electrons and the direct deposition of charge. The angular and energy distributions of the emitted secondary electrons are crucial for understanding the net charging rate. For example, the angular distribution of secondary electrons escaping from a gold surface under a 1 keV primary electron beam closely follows a cosine function peaking around 45° and exhibiting near inversion symmetry ([figure 6i](#)) [8]. The energy distribution of low-energy electrons (LEE) produced by GCR interactions is typically narrow, centered around 8.4 eV with a standard deviation of 2 eV and their emission direction is generally assumed to be isotropic relative to the surface normal. These characteristics influence both the magnitude and variability of the charge deposited on the TMs. The yield of secondary electron emission depends on the energy and type of the incident GCR particle. For protons, the yield can be parameterized using empirical fits ([eqn. 1](#)), such as $Y_0 = 2.18$, $E_0 = 180$ keV and $s = 1.54$, while for alpha particles, the yield is lower and less significant compared to protons, with $Y_0 = 6$, $E_0 = 200$ keV and $s = 1.64$ ([figure 6ii](#)) [18].

$$\text{No. of electrons generated per incident GCR particle: } Y(E, \theta) = Y_0 [1 + 0.66 \cos^{0.8} \theta] \frac{sE/E_0 [1 - 0.7(1 - \cos \theta)]}{s - 1 + (E/E_0 [1 - 0.7(1 - \cos \theta)])^2} \quad (1)$$

The interaction of GCRs with the spacecraft is not limited to direct charging of the TMs. Secondary processes such as the generation of Cherenkov X-rays in surrounding materials can also contribute to the overall charge environment. The number of Cherenkov photons emitted depends on the energy and trajectory of the incident particle as well as the properties of the medium and can be described by specific yield equations that account for the photon emission angle and the complex formation zone of the material [6].

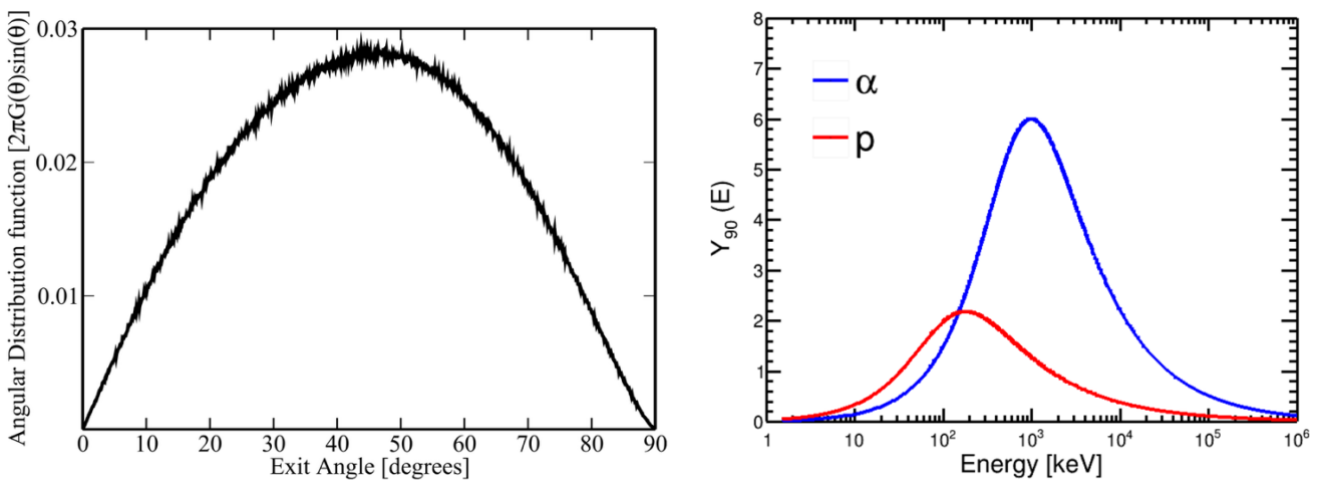


Figure 6: i) Angular distribution of secondary electrons escaping gold surface for 1 keV primary electron beam [8] ii) Low-energy electron yield of the kinetic emission processes for proton (red) and alpha particles (blue) [18]

6.1.2 Solar Energetic Particles (SEPs)

SEPs are a significant source of charging with their influence being especially pronounced during periods of heightened solar activity. SEPs are primarily accelerated during solar eruptive events such as coronal mass ejections CMEs (coronal plasma and magnetic field lines are ejected into interplanetary space) and solar flares which inject large fluxes of high-energy protons and heavier ions into the heliosphere. The temporal and spatial characteristics of SEP events are closely linked to the dynamics of the solar magnetic field and the geometry of the CME shock front. The most active phase of the solar cycle which lasts approximately seven years, two years preceding the solar maximum and about four years following it correlates with the highest probability of intense SEP events. The propagation of SEPs from the Sun to the Earth is not isotropic, the spiral structure of the solar magnetic field introduces an asymmetry in the intensity-time profiles of SEP fluxes. Events originating from the western hemisphere of the Sun are more likely to result in SEPs reaching the Earth and by extension, spacecraft such as LISA PF, compared to those from the eastern hemisphere. This is due to the configuration of the interplanetary magnetic field which channels particles more efficiently from western longitudes. The shock structure of a CME can be divided into three regions: the eastern and western flanks and the central shock nose ([Figure 7](#)). The central region or shock nose, is characterized by a compression of magnetic field lines, which can further enhance particle acceleration. The dynamics of SEP propagation and the resulting fluxes at the spacecraft are therefore highly dependent on the CME's origin and evolution. For western events, the magnetic connectivity to the Earth is optimal, leading to higher SEP fluxes and a greater likelihood of significant charging events on the proof masses [19].

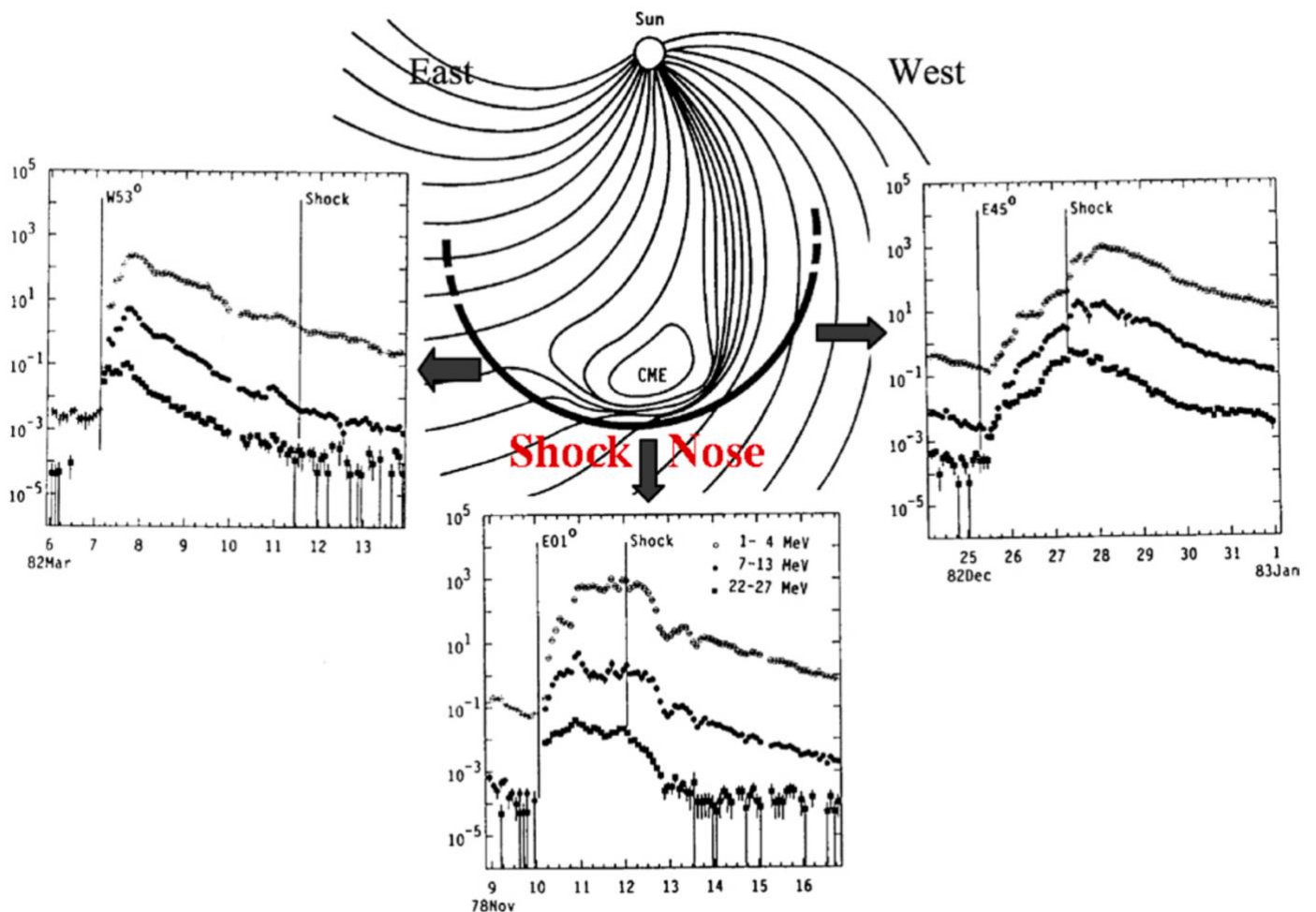


Figure 7: Illustration of a CME's shock structure showing the central shock nose and its eastern and western flanks propagating through the interplanetary magnetic field. The graphs show typical SEP intensity-time profiles demonstrating how the particle flux observed at Earth varies significantly depending on the event's solar longitude [19]

The energy spectrum of SEPs is broad but for the purposes of charging, the most relevant particles are those with energies sufficient to penetrate the spacecraft shielding and reach the TMs. Protons

and ^4He nuclei are the dominant species in SEPs and their energy distributions can be described by power-law spectra with parameters that vary depending on the solar event and the phase of the solar cycle. The ability of these particles to penetrate the shielding is energy-dependent, for instance, protons and ^4He nuclei with energies below 100 MeV/nucleon are generally unable to reach the TM due to the attenuation provided by the surrounding material. Only higher-energy particles contribute directly to charging by depositing charge or inducing secondary electron emission upon interaction with the TM or its housing.

6.1.3 Triboelectric effects

The triboelectric effect is associated with the transfer of charge due to frictional contact or relative motion between different materials. In the LISA PF environment, triboelectric charging may occur during launch, handling or in the presence of micro vibrations and mechanical contacts between the TM and its housing or other spacecraft components. Although the operational phase of LISA PF is designed to minimize such contacts, any residual triboelectric charge acquired during earlier mission phases can persist due to the extremely low conductivity and high vacuum conditions in space. The presence of even small triboelectric charges can introduce spurious forces on the TMs, necessitating careful pre-launch procedures and in-flight charge management strategies. This effect can be activated also due to rapid changes in the local radiation environment caused by the passage of interplanetary coronal mass ejections or high-speed solar wind streams [20].

6.1.4 Gamma Ray Bursts

During the time LISA PF remained in orbit, no solar energetic particle events or major astrophysical phenomena were observed, but gamma-ray bursts (GRBs) that can cause TM charging exist. These events are among the most extreme astrophysical phenomena, characterized by intense bursts of hard X-rays and gamma rays that can last from milliseconds to several minutes. They are generally classified as either short or long bursts. Short GRBs with a duration of less than two seconds are understood to originate from the merger of compact objects like two neutron stars or a neutron star and a black hole. Long GRBs lasting longer than two seconds are associated with the core collapse of massive stars. A special class of these events are magnetar flares which are the brightest phenomena in our galaxy and are considered a type of short GRB.

The charging of the TMs by these major astrophysical phenomena presents a unique signature when compared to the charging caused by GCRs and SEPs. While GCRs and SEPs charge the LISA PF TM positively, photons from GRBs and magnetar flares induce a negative charge. This abrupt change in the net TM charging from positive to negative, combined with a sudden increase in the charging noise would serve as a distinct indicator of a GRB or magnetar flare. Simulations of intense events such as the long GRB 221009A show that the resulting change in net charging and the increase in charging noise would be clearly observable above the background GCR levels. It is estimated that even during these extreme events the increase in TM charging noise would remain below the mission's required sensitivity, allowing for the potential of simultaneous GRB and gravitational wave detection [21].

6.2 Charging Mechanism

When particles penetrate the spacecraft shielding, they can directly ionize the proof mass or surrounding materials, leading to the emission of secondary electrons and the generation of electron-hole pairs. The resulting cascade of secondary electrons is a significant contributor to the overall charge transfer as these low-energy electrons can escape from surfaces if their kinetic energy perpendicular to the boundary exceeds the material's work function, a process that is especially relevant for metallic surfaces [6]. The secondary electron yield exhibits a soft energy spectrum, peaking at a few electron volts and is sensitive to the incident particle's energy and the material properties [8]. Hadronic interactions also play a role in secondary particle production. When high-energy hadrons interact with nuclei in the spacecraft materials, they can induce nuclear reactions leading to the emission of neutrons, protons, alpha particles and heavier nuclear fragments. These processes are modeled using a combination of elastic and inelastic scattering,

with inelastic reactions further subdivided into evaporation, gamma emission and fission for heavier nuclei. The production of secondary hadrons is enhanced by photonuclear reactions which can be initiated by high-energy photons, electrons or muons. The resulting nuclear fragments and secondary particles can themselves contribute to further ionization and electron emission, compounding the overall charging effect [22].

In addition to direct ionization and secondary electron emission, backscattering of primary particles at the proof mass or electrode surfaces can also contribute to charge transfer. The reflection yield quantifies the fraction of incident primaries that are backscattered. The backscattered primaries, together with the secondary electrons establish a dynamic equilibrium of charge exchange between the proof mass and its environment [20]. Electrostatic patch potentials on the electrode surfaces introduce further complexity to the charge transfer mechanisms. These patch potentials couple to the net free charge on the TM, generating additional acceleration noise that must be carefully managed [9].

The deposition of charge is determined by the stopping points of both primary and secondary particles within the TM. Results in [11] indicate that primary protons upon entering the TM exhibit a relatively broad spatial distribution of stopping points with a tendency to penetrate deeper into the mass. In contrast, secondary electrons which are produced by ionization processes initiated by the primaries tend to deposit their energy and thus their charge much closer to the surface of the TM. This leads to a spatial gradient in charge density with higher concentrations near the surfaces exposed to the incident flux. The two-dimensional projections of these distributions reveal that the charge is not homogeneously spread but is instead localized particularly for the secondary electrons. The temporal evolution of charge accumulation is closely linked to the variability of the incident particle environment. During periods of quiescent GCR flux the charging rate is relatively stable, leading to a slow and steady build-up of charge. However, during SEP events the incident flux can increase by several orders of magnitude within minutes to hours, resulting in rapid and substantial changes in the net charge on the TMs [16].

6.3 Consequences of Test Mass charging

Electrostatic force noise arises from both intrinsic and extrinsic sources associated with the charging and discharging processes of the TMs and their surrounding electrodes. One of the primary contributors to electrostatic force noise is the stochastic nature of charge deposition on the TMs which is driven by the interaction with CGRs and SEPs. The resulting charging current exhibits shot noise, characterized by a spectral density that is proportional to the square root of the effective charging rate λ_{eff} and the elementary charge e . This relationship can be expressed as $S = \sqrt{2e^2\lambda_{eff}}$, directly linking the fluctuations in charging to the force noise experienced by the TMs. The impact of electrostatic force noise on measurement sensitivity is particularly critical in the low frequency regime that will be targeted by LISA, where the acceleration noise requirements are most stringent [10]. Electrostatic noise is further exacerbated by the presence of fluctuating patch potentials and surface contamination. The work function of gold, commonly used for the TM surfaces, can vary significantly due to adsorption of contaminants such as carbon and oxygen, leading to local variations in surface potential. These variations induce additional force noise as the TM moves relative to the surrounding electrodes and their temporal fluctuations can couple into the measurement band, degrading sensitivity [8].

Acceleration noise in the context of LISA PF arises predominantly from the interaction of the TMs with their environment, where charging phenomena play a central role. These charged-induced forces manifest as acceleration noise which can obscure or mimic the signals LISA PF is designed to detect [19]. The primary mechanism by which charge accumulation leads to acceleration noise is through the generation of Coulomb and Lorentz forces. When a TM acquires net charge, it interacts electrostatically with the surrounding electrode housing and with fluctuating electric fields resulting in Coulomb forces. Simultaneously, as the charged TM moves through ambient magnetic fields, Lorentz forces are induced. Both types of forces can introduce spurious accelerations thereby

contaminating the measurement bandwidth. The magnitude of these disturbances is not static, it depends on the instantaneous charge state of the TM, the geometry of the electrode housing and the local electromagnetic environment [23]. Coherent noise components also arise due to the time-dependent nature of the TM charge. As the charge on the TM evolves, it can produce periodic signals within the measurement bandwidth. These coherent components can become substantial, especially if the TM is not discharged frequently enough. The amplitude of these signals is influenced by the rate of charge accumulation and the specific design of the sensor including the mass of the TM and the gap between the TM and the electrodes [24]. Increasing the gap can reduce Coulomb disturbances but this comes at the cost of decreased sensor sensitivity, necessitating a careful balance in the design [22].

7 Mitigation strategies for charging of proof masses

7.1 Active charge control techniques

7.1.1 Use of UV Light

The LISA PF mission incorporated a UV discharge system to mitigate TM charging. This system operated on the principle of the photoelectric effect by offering a non-contact method to discharge the TMs. For the gold-coated surfaces of the TMs and the surrounding electrode housing, UV light with a wavelength $\leq 239 \text{ nm}$ (or $\leq 289 \text{ nm}$ for contaminated surfaces) was required to provide sufficient energy to overcome the material's work function (5.2 eV for pure Au, or $\geq 4.3 \pm 0.4 \text{ eV}$ for air-exposed Au) facilitating the emission of electrons. LISA PF primarily utilized Hg lamps in a DC intermittent discharge mode which involved shining UV light continuously every 2-3 weeks to discharge the TM, thereby necessitating a pause in scientific observations.

For the upcoming LISA mission the Charge Management System will employ Ultraviolet Light Emitting Diodes (UV LEDs) which will provide advanced pulsed charge control method which synchronizes UV pulses with an existing 100 kHz high frequency electric field within the gravitational reference sensor. This synchronization is crucial for facilitating the desired direction of photoelectron current, removing the need for applied DC voltages to guide the charge current and offering greater control over electron movement. This capability limits shot noise and TM acceleration noise. The effectiveness of this system is understood through the Apparent Yield (AY) curve ([Figure 8](#)) which characterizes the rate of TM charge per UV light power ($\dot{q}/P_{UV}$) as a function of the TM potential (V_{TM}). The TM reaches an equilibrium point when the AY is zero, indicating no net photoelectron current. An analytical model for charge control describes the electron discharge rate $\dot{q}_{CC}$ which when linearly approximated near equilibrium can be expressed as $\dot{q}_{CC} = \alpha(q_{CC} + q_{ac} - q_{eq})$. Here $\alpha = \beta P/C_T$ where P is the UV power (in photons/s), C_T is the total TM capacitance and β is the constant slope of the near-linear AY function.

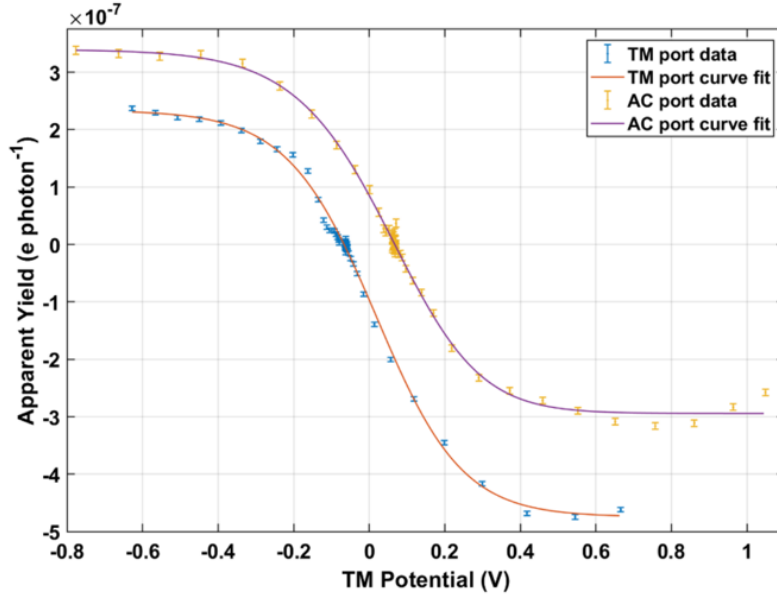


Figure 8: measured UV light AY curves using TM and AC injection ports with their fitted empirical models [25]

The solution to this differential equation shows that the TM charge q_{CC} decays exponentially over time towards a new equilibrium $q_{CC} = (q_0 - q_{eq} + q_{ac})e^{\alpha t} + q_{eq} - q_{ac}$. In this equation q_{ac} represents an additional charge contribution due to the pulsing UV light while q_{eq} is the intrinsic equilibrium charge, indicating that precise control over the TM's equilibrium point is achievable through pulsed charge control by adjusting q_{ac} . When considering the influence of the space radiation environment represented by a constant environmental charging rate R the equation for the total TM charge q simplifies to $q = (q_0 - q'_{eq})e^{\alpha t} + q'_{eq}$ where $q'_{eq} = q_{eq} - q_{ac} - \frac{R}{\alpha}$. This implies that if q_{ac} is set to $q_{eq} - \frac{R}{\alpha}$ the desired equilibrium voltage can be zero [25].

7.1.2 Discharge approaches

Periodic/Intermittent Discharge: This method involves allowing the LISA PF TMs to accumulate charge over several weeks, typically drifting between -70 mV and 70 mV, before being rapidly discharged using low pressure mercury (Hg) lamps. While effective in managing charge, this approach required pausing science mode during discharge. The rapid discharge caused a increase in charge noise (up to 100 times higher than cosmic ray noise) due to the stochastic release of approximately 30,000 elementary charges per second. Furthermore, the time-dependent charge accumulation created unacceptable quasi-periodic "saw-tooth" Fourier components in the science data which could exceed noise limits by a factor of 20 and TM potentials beyond ± 70 mV could violate mission acceleration noise requirements.

Continuous Discharge: In contrast, continuous discharge aims to maintain a constant TM potential by continuously balancing the natural charging with a steady, controlled discharge rate, thereby allowing uninterrupted science operations. Simulations demonstrate that this pulsed continuous mode maintains the TM potential very close to 3 mV and significantly reduces force noise, outperforming LISA PF's intermittent mode by a factor of 10 and falling 50 times below the overall LISA noise budget. Even at lower UV power levels this method can effectively suppress environmental charge and maintain TM potential well below the 70 mV limit over long durations (e.g. 500 days). Although continuous discharge introduces its own shot noise, increasing the discharge rate to manage variations in Galactic Cosmic Ray (GCR) flux is tolerable in terms of acceleration noise penalties.

7.1.3 Low energy electron irradiation

Low energy electron irradiation presents an alternative solution for neutralising electrostatic charges on TMs, specifically designed to be compatible with cryogenic temperatures. This method relies on a material's intrinsic response to electron irradiation, defined by its Secondary Electron Yield (SEY)

which is the ratio between the number of emitted secondary electrons and the number of incident primary electrons. Depending on the incident electron energy, SEY can be lower, equal to or greater than 1, allowing for the addition or removal of electrons from a surface until charge neutralisation is achieved. For implementation, an electron gun irradiates an electrically insulated sample at selected energies and the induced surface voltage is measured by a non-contact electrostatic voltmeter. The SEY is quantitatively determined by the equation $SEY = \frac{I_{out}}{I_p} = 1 - \frac{I_s}{I_p}$, where I_p is the primary electron beam current and I_s is the sample drain current to ground. Experiments show that positive charging occurs when $SEY > 1$ as the number of emitted electrons I_{out} exceeds the incident electrons I_{in} , leaving a net positive charge on the surface. Conversely, negative charging occurs when $SEY < 1$ as $I_{out} < I_{in}$ meaning electrons accumulate on the surface. Neutralisation is achieved when $SEY = 1$, at which point the number of incident and emitted electrons are equal $I_{in} = I_{out}$ causing the surface to balance excess charges and reach a stable surface voltage of zero $V_s = 0$. This method offers several advantages including its compatibility with cryogenic environments, the ability to compensate for charges of both polarities, minimal expected effects on optical surfaces due to low electron penetration and a potential synergistic use for defrosting by inducing molecular ice non-thermal desorption [26].

7.2 Passive mitigation approaches

7.2.1 Structural Shielding

To ensure the high level of purity required for the geodesic motion of TMs along the interferometer axes, the caging mechanism features in the LISA Technology Package for LISA PF are strategically positioned along the z-axes. This placement is crucial because the x-axis is reserved for the interferometric measurements. By aligning the caging mechanism features along the z-axis, disturbances caused by charging that would otherwise be significant along the sensitive measurement axes (x and y) are minimized. This implementation helps to limit acceleration noise along the measurement axis. Moreover, to minimize electrostatic disturbances, caging mechanism components have rounded edges and the stoppers are positioned at a minimal height. Two stopper designs were evaluated: a "baseball-glove" shape and an elongated one (Figure 9). Finite element analysis showed the elongated stopper produced significantly higher capacitance and electrostatic disturbances. As a result the baseball-glove design was selected for the LISA Test Package due to its superior electrostatic performance. Furthermore, A symmetric design of the caging mechanism is also crucial for canceling unwanted electrostatic forces. These stray accelerations a_Q are driven by gradients in capacitance with respect to displacement (k). The force is proportional to the gradient $a_Q = \partial C / \partial k$. In a perfectly symmetric system with a centered TM this capacitance gradient vanishes, nullifying the force. For the LISA Test Package, this was achieved by implementing a pyramidal plunger head and adding a second set of stoppers [24].

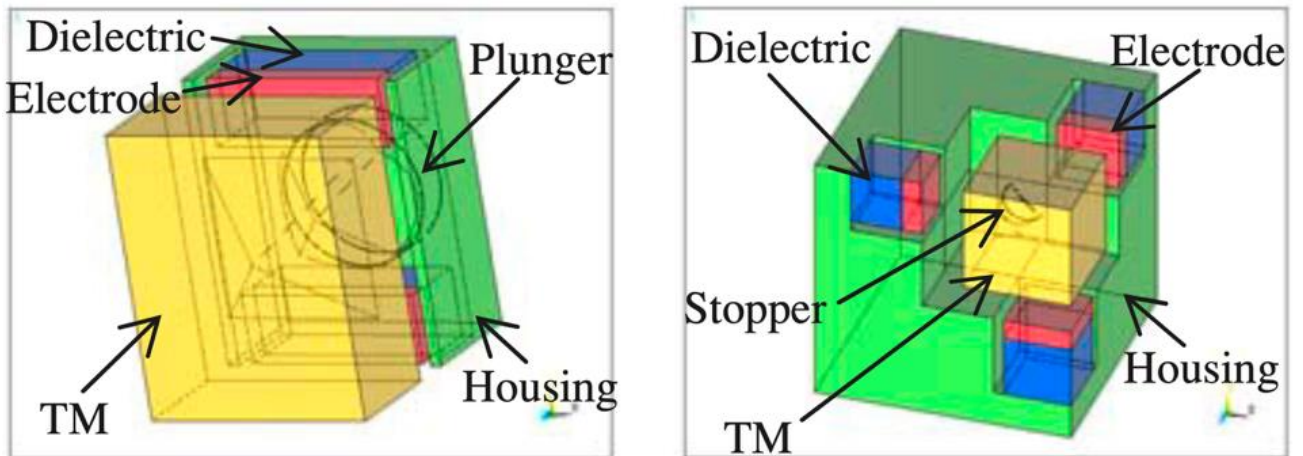


Figure 9: Pyramidal plunger and baseball-glove stopper [24]

7.2.2 Electrical Configuration

For the electrical configuration of LISA PF inertial sensors, several strategies are employed to mitigate the effects of TM charging. Minimizing sensor voltage and geometrical offsets is a critical approach as spurious forces on the TM arise from the interaction of accumulated charge with surrounding conducting surfaces and applied potentials. The resulting acceleration can be approximated by terms dependent on voltage difference δ_V and displacement δ_d as $a \approx \frac{C_i \delta_V}{m d_i} \left(V_i - \frac{\bar{Q}}{C} \right) + \frac{2 C_i \delta_V}{m d_i^2} \left(V_i - \frac{\bar{Q}}{C} \right)^2$ where C_i is the capacitance between TM and i^{th} electrode, d_i is nominal distance between TM and electrode, V_i is DC potential of that electrode, $\bar{Q}$ is the average charge on TM and C is the total capacitance.

Reducing these offsets for example by achieving voltage differences of 10 mV on conductors along different axes can significantly decrease coherent signals, stiffness and overall noise predictions [22]. Guard rings are incorporated into the sensor design to limit fringing fields and unwanted cross-couplings between different elements and degrees of freedom [24]. Furthermore, a grounding protection ring is formed by grounding the area between sensing and injection electrodes to effectively suppress direct coupling, ensuring electric field lines terminate at the TM surfaces and improving sensitivity. Lastly, electrical isolation of electrodes typically achieved using ceramic bushings and forming a complete Faraday cage structure with the housing shields the TM from external electromagnetic interference. Beyond these design features, insulating spacer placement and TM-electrode separation are also crucial for managing charging disturbances. Insulating spacers are carefully positioned within electrode coverage to prevent exposure to the TM's field of view, thereby avoiding unwanted interference and maintaining signal reliability. Increasing the nominal separation between the TM and electrodes can lead to significant reductions in Coulomb disturbances as increasing x-face separations from 4 mm to 6 mm could reduce these disturbances by 50% to 70%. However, this benefit must be balanced against a potential decrease in sensor sensitivity. In LISA PF the sensitive x-axis has a 4 mm sensing gap specifically designed to mitigate electrostatic stiffness [9].

7.3 Charge measurement and control

7.3.1 Measurement and monitoring

LISA PF incorporates radiation monitors (RMs) ([Figure 10](#)) to actively survey the interplanetary space environment particularly GCRs and Solar Energetic Particles SEPs which are known to induce charge buildup on the TMs. These monitors are designed to measure integral fluxes of galactic and solar particles with energies exceeding 75 MeV/n. Functionally, the RMs provide counting rates of particles traversing their silicon layers and measure ionization energy losses for particles in coincidence mode. This capability enables them to detect and identify low-fluence SEPs and monitor low frequency variations in cosmic ray flux, offering crucial insights into space weather and solar physics investigations. The LISA PF RM consists of two 300 μm thick silicon PIN photodiodes each with a 147 mm^2 area, arranged in a telescopic configuration within a 6.4 mm thick copper enclosure. This setup oriented along the Sun-Earth direction ensures shielding from particles below approximately 70-100 MeV/n. Data from the RM, stored as histograms over 600-second periods, have shown a reliable correlation with the test-mass charging rate [16].

Furthermore, TM charge measurement on LISA PF is critical for evaluating parasitic electrostatic forces. This process involves actively determining the charge level and its sign as well as characterizing any residual charge on the TMs post release. The measurement technique relies on inducing a force or torque on the test mass that is proportional to its potential V_{TM} or charge q . This is achieved by applying modulating voltages V_{MOD} with opposite phases to sensing electrodes within the Gravitational Reference Sensor. The resulting force F_x along the x-axis or torque Γ_ϕ rotation can be described by the following crucial equations $F_x = -4 V_{MOD} \frac{q}{C_T} \left| \frac{\partial C_x}{\partial x} \right| \sin 2\pi f_{MOD}$ and $\Gamma_\phi =$

$-4 V_{MOD} \frac{q}{C_T} \left| \frac{\partial C_x}{\partial \phi} \right| \sin 2\pi f_{MOD}$. Here C_T represents the total capacitance of the test mass, $\frac{\partial C_x}{\partial x}$ and $\frac{\partial C_x}{\partial \phi}$ are derivatives of capacitance and f_{MOD} is the modulation frequency. The TM response is sensed via the capacitive sensor system or for higher sensitivity by the on board laser interferometer and then demodulated at f_{MOD} to calculate the charge. These continuous measurements allow for studying the time dependence of charging and estimating the charging rate through linear fitting [7].

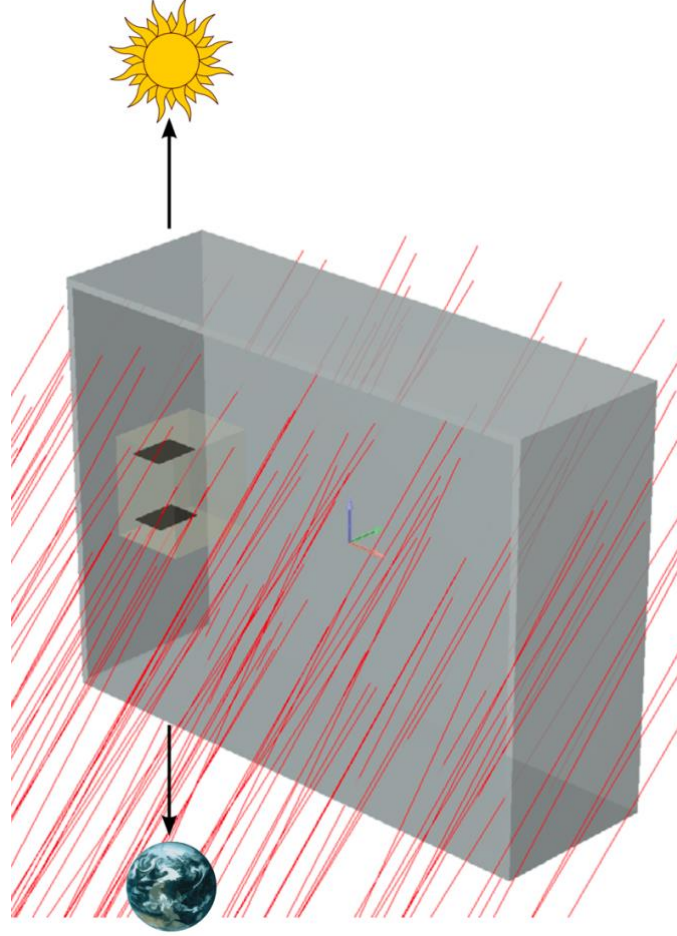


Figure 10: A model illustrating the orientation of the LISA-PF radiation monitor, with lines representing the simulated trajectories of incident solar protons from the 23 February 1956 SEP event [16]

7.3.2 Charge neutralisation

Upon initial uncaging the LISA PF TM is subject to triboelectric charging which results in an unknown residual charge, typically negative. Measurements confirmed initial TM potentials ranged from -12 to -512 mV [7]. This initial charge must be nullified as it can exceed the approximately 100 mV operational limit. To achieve this the charge control system first measures the TM's charge and sign. Subsequently, UV radiation is directed onto specific surfaces: the surrounding electrode housing if the TM is positively charged or the TM itself if negatively charged to release photoelectrons and reduce the charge. This process is repeated iteratively until the charge is brought within acceptable limits for science operations [27]. The fundamental relationship between TM potential V_{TM} and charge q is given by $V_{TM} = q/C_T$ where C_T is the total capacitance of the TM to ground making monitoring the potential equivalent to monitoring the charge [7].

Beyond direct charge removal the mitigation of charge induced force noise involves careful management and compensation of electric fields. Electrostatic force noise can arise from the coupling of a fluctuating TM charge $\delta q(t)$ with a mean DC electric field $\Delta_{o,x}$ or a fluctuating electric field $\delta x(t)$ with the mean TM charge q_o . To reduce the impact of the former, the inherent electric field $\Delta_{o,x}$ within the gravitational reference sensor can be measured and a counter-field applied via the electrodes. This directly addresses the force component $F_x = -\frac{q}{C_T} \left| \frac{\partial C_x}{\partial x} \right| \Delta_x$ [25]. Furthermore,

experiments have demonstrated that applied DC voltages to electrodes can actively bias the TM to a neutral potential, influencing the environmental charging rate by managing the flow of this secondary electron cloud, steering it to land on or off the TM using injection electrodes (Figure 11) as needed to keep the TM electrically neutral. Even the amplitudes of AC actuation voltages can affect the charging rate likely due to their influence on electron emission and charge transfer between the electrode housing and the TM [7]. Moreover, Lorentz forces on a charged TM moving through magnetic fields are mitigated by a metallic enclosure creating a counter-balancing internal electric field [27].

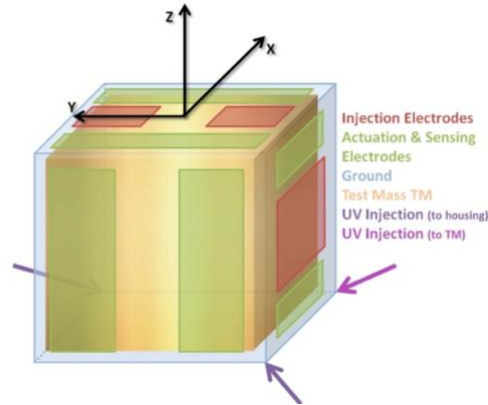


Figure 11: LISA PF gravitational reference sensor orientation and injection electrodes to remove electrons plus control them [25]

7.4 Novel Methods

7.4.1 Mirror based discharge

Mirror based discharge (Figure 12i) is a mitigation strategy designed to prevent high level negative spacecraft charging by enhancing photoemission from a spacecraft's dark or shadowed surfaces. The core process involves reflecting sunlight onto these dark-side surface elements which then generate photoemission currents. This increased photoemission current I_{ph} significantly exceeds the incoming ambient electron flux thereby making negative charging impossible. Consequently, this method effectively raises the critical temperature for the onset of charging, making it more difficult for the spacecraft to accumulate negative voltage. Implementation involves using aluminum mirrors which exhibit high reflectance, even in the vacuum ultraviolet (VUV) range. If a spacecraft features a combination of conducting and dielectric surfaces on its dark side the reflected light should be focused on the dielectric areas for effective mitigation. The photoelectron current generated by this reflected light can be calculated as $I = \int_0^\infty dE R(E)f(E)\gamma(E)$ where $R(E)$ is the reflectance, $f(E)$ is the solar photon spectrum and $\gamma(E)$ is the photoelectron yield function. However, since highly reflective mirrors emit almost no photoelectrons, they can charge negatively themselves; this can be mitigated by positioning them far from the satellite, attaching a sunlit conducting foil, or incorporating sharp spikes to shed excess electrons.

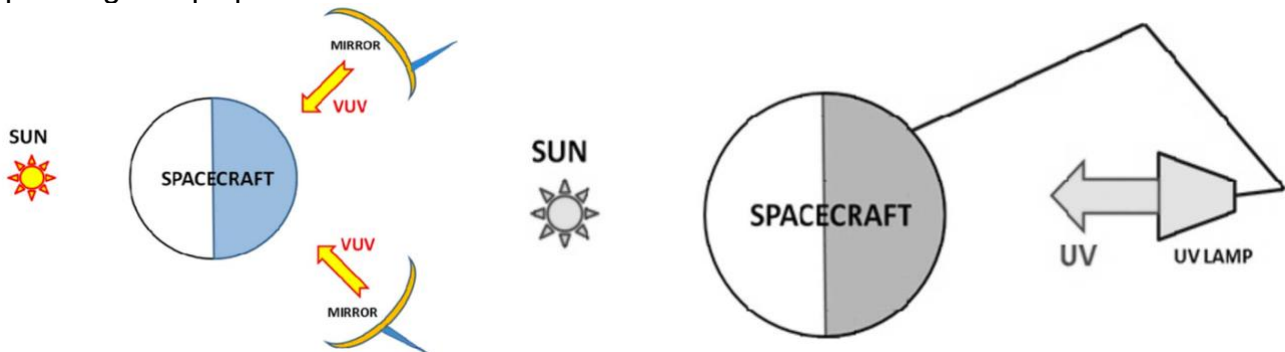


Figure 12: i) Mirror based discharge ii) UV Lamp [15]

7.4.2 Artificial UV/VUV Photon Sources

Artificial UV/VUV photon sources ([Figure 12ii](#)) offer a novel mitigation strategy to prevent high-level negative voltage charging on spacecraft also on their dark sides and during eclipse. These sources, including xenon excimer lamps, VUV LEDs and deuterium UV lamps operate by shining artificial ultraviolet or vacuum ultraviolet (VUV) light onto shadowed spacecraft surfaces to generate photoemission. The VUV photons must possess sufficient energy to overcome the material's work function (typically 4–5 eV). This process ensures that the generated photoelectron currents substantially exceed the incoming ambient electron flux, thereby effectively mitigating negative charging. The photoelectron current (I) generated by these VUV lamps can be calculated using the equation $I = \int_0^\infty dE f_L(E)Y(E)$ where $f_L(E)$ represents the photon spectrum of the VUV lamp and $Y(E)$ is the photoelectron yield function of the surface material for each incoming photon of energy E . For implementation, these lamps are positioned and oriented to produce fairly even VUV illumination across surfaces and can be supported by metal rods that allow for rotation. To prevent the VUV lamps themselves from charging to high negative voltages, designers employ strategies such as all-metal surface construction or by extending a light-weight metallic plate or foil from the device or installing sharp spikes on the back of the metallic devices [15].

7.4.3 Machine Learning (ML) models

ML models are used to distinguish between recurrent and non-recurrent GCR variations which is crucial for analyzing radiation data and estimating TM charging. These are an ensemble of four convolutional neural networks (CNNs) each with a unique design (supervised ML models). These CNNs use several inputs for their predictions:

- Hourly solar wind speed (over 9 days) and interplanetary magnetic field intensity (over 3 days)
- Statistical measures (min, max, median, and trends) derived from these time series and their sub-windows
- The GCR value from nine days before the prediction as previous GCR status affects new variations

The CNNs include convolutional layers (e.g. 64 filters of size 3 or 2) and fully connected layers (e.g. 32 and 16 neurons or 128 neurons), utilizing activation functions like rectified linear (ReLU) or hyperbolic tangent. They are trained on hourly data from LISA Pathfinder (LPF) and ACE missions (2016-2017) with a 10% validation set for monitoring performance and early stopping. The main goal is to identify interplanetary structures like high-speed solar wind streams (HSS) and interplanetary coronal mass ejections (ICME) as these affect GCR flux and thus TM charging differently. The models successfully predict recurrent GCR variations (linked to HSS) but fail to predict non-recurrent Forbush Decreases (caused by ICME) [20].

8 Visualising using COMSOL Multiphysics

Sections 6 and 7 established the theoretical challenge of TM charging and the necessity of mitigation strategies. This section now provides visualization of the spurious forces at the heart of this problem using finite element modelling, demonstrating why this mitigation is essential.

8.1 COMSOL model explained

A simplified 2D as shown in [Figure 13](#) model was developed using COMSOL Multiphysics software. While analytical formulas provide force estimates, a finite element model offers a powerful visual tool to see how electric fields and charges behave in a realistic geometry. The goal of this simulation is to create a clear visualization of how small and unavoidable imperfections such as a slight misalignment of the TM, a net electric charge, and non-uniform surface potentials combine to create disturbing forces at the femto-Newton level.

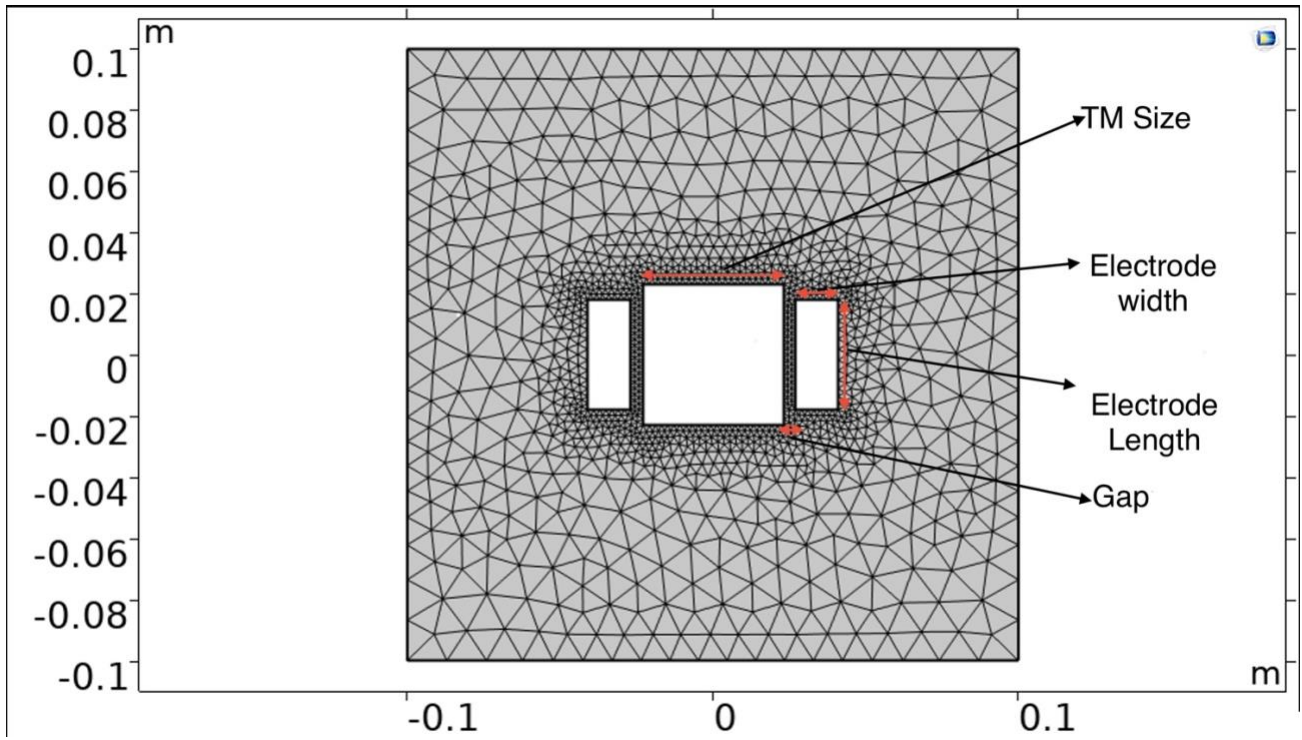


Figure 13: Geometry and Mesh

8.1.1 Parameters, Geometry & Materials

Parameter	Value
TM Size	46 mm
Gap	4 mm
Electrode length	36 mm
Electrode width	14 mm
Offset in x direction	0.2 mm

Table 1: Parameters and their values

A 2D cross-section of the TM inside its electrode housing was created. The primary spurious force in LPF is along the sensitive x-axis. A 2D cross-section is an effective simplification that allows us to focus on the interaction between the TM and the x-axis electrodes without the computational expense of a full 3D model. The TM, the left and right electrodes were drawn (as voids) and a large outer box representing the vacuum. Inside an ideal conductor in an electrostatic state, the electric field is zero. The software doesn't need to calculate anything for the volumes of the TM and electrodes, it only needs to solve for the complex electric field in the vacuum gaps and at boundaries. This method creates a single computational domain (the vacuum material) and defines the conductor surfaces as internal boundaries. This is the standard and efficient method for electrostatic simulations as it tells the software to only solve for the electric field in the space between the conductors where the physics is happening [9]. The dimension values used for the components is given in [Table 1](#).

8.1.2 Physics & Studies

For Force Calculation: The primary goal is to simulate the physical scenario described in the literature where a charged, isolated proof mass is influenced by stray electric fields from its housing. Boundary conditions from the Electrostatics interface was used to model this specific situation. A Floating Potential condition with a total charge of $10^7 e$ (a typical value for LISA PF [28]) was applied to the TM boundaries. This correctly models an isolated conductor with a net charge. An Electric Potential of 0 V was applied to the left electrode to act as a ground reference. To simulate patch potentials the right electrode's inner face was partitioned. Each segment was assigned a different small Electric Potential (e.g +20 mV, -15 mV). This allows for the direct calculation of the resulting spurious force due to a net TM charge, its off center position and non-ideal electrode surface

potential [29] [30]. A parameter sweep of the offset was also done to show variation of forces at different misalignments.

For Capacitance Calculation: To validate that the 2D model has the correct fundamental electrostatic properties calculating its capacitance allows for a direct comparison with the known capacitance values of the real 3D GRS ensuring our model is a credible representation. Terminal boundary conditions were applied to the TM and both electrodes. A Stationary Source Sweep study was used which is COMSOL's built-in tool for energizing each terminal individually to compute the complete capacitance matrix of the system. This method provides a direct check of the model's accuracy. If the simulated capacitance is realistic we can have high confidence that the simulated forces and fields are also realistic.

8.2 Results

8.2.1 Force

The primary simulation goal was to quantify the spurious electrostatic force acting on the TM. This force is generated by the complex interaction between the TM's net charge and the asymmetric electric fields from its housing. The simulation was configured to represent a semi-realistic non-ideal scenario. The TM was modeled as a floating potential with a net charge of $10^7 e$. Asymmetries were introduced in two ways:

- Geometric Asymmetry: The TM was given a lateral offset from the central position (a standard case of 0.2 mm was used)
- Electrical Asymmetry: Millivolt-level patch potentials (+20 mV and -15 mV) were applied to segments of the right electrode to simulate a non-uniform surface while the left electrode was held at 0 V

For this standard case (0.2 mm offset) the model calculates a net spurious force in the sensitive x-direction. The force per unit length (N/m) is integrated over the four boundaries of the TM and multiplied by the electrode length (36 mm) to find the total force. The resulting simulated force is $0.848 \times 10^{-13} \text{ N}$ in x-direction as in [Table 2](#) exported from COMSOL.

Force in x-direction (N)	Force in y-direction (N)
0.848×10^{-13}	-8.284×10^{-17}

Table 2: Exported derived values of Forces in x and y direction

This simulated result shows strong agreement with analytical estimates derived from LISA Pathfinder flight data. In [28], the spurious force F is described by the relationship:

$$F_x = \frac{q}{C_T} \left| \frac{\partial C_x}{\partial x} \right| \Delta_x$$

With TM charge $q = 10^7 e$, Total Capacitance $C_T = 34.2 \text{ pF}$, gradient of x – axis capacitance $\left| \frac{\partial C_x}{\partial x} \right| = 291 \frac{\text{pF}}{\text{m}}$ and potential asymmetry $\Delta_x = 20 \text{ mV}$. The analytical force value is $2.7228 \times 10^{-13} \text{ N}$.

Our simulated value of $0.848 \times 10^{-13} \text{ N}$ is in the same femto-Newton range and is the same order of magnitude as the analytical value derived from flight data. The difference between the values is expected as our 2D model is a simplification of the full 3D geometry and uses an estimated distribution of patch potentials on one of the electrodes. The strong agreement in magnitude validates that the model correctly captures the physics of the force. We also see an extremely small force of $8.284 \times 10^{-17} \text{ N}$ in the downward y-direction.

To further investigate the role of geometric asymmetry, a parametric sweep was performed varying the TM's lateral offset from -0.5 mm to +0.5 mm in steps of 0.1 mm. The resulting forces are presented in [Figure 14](#).

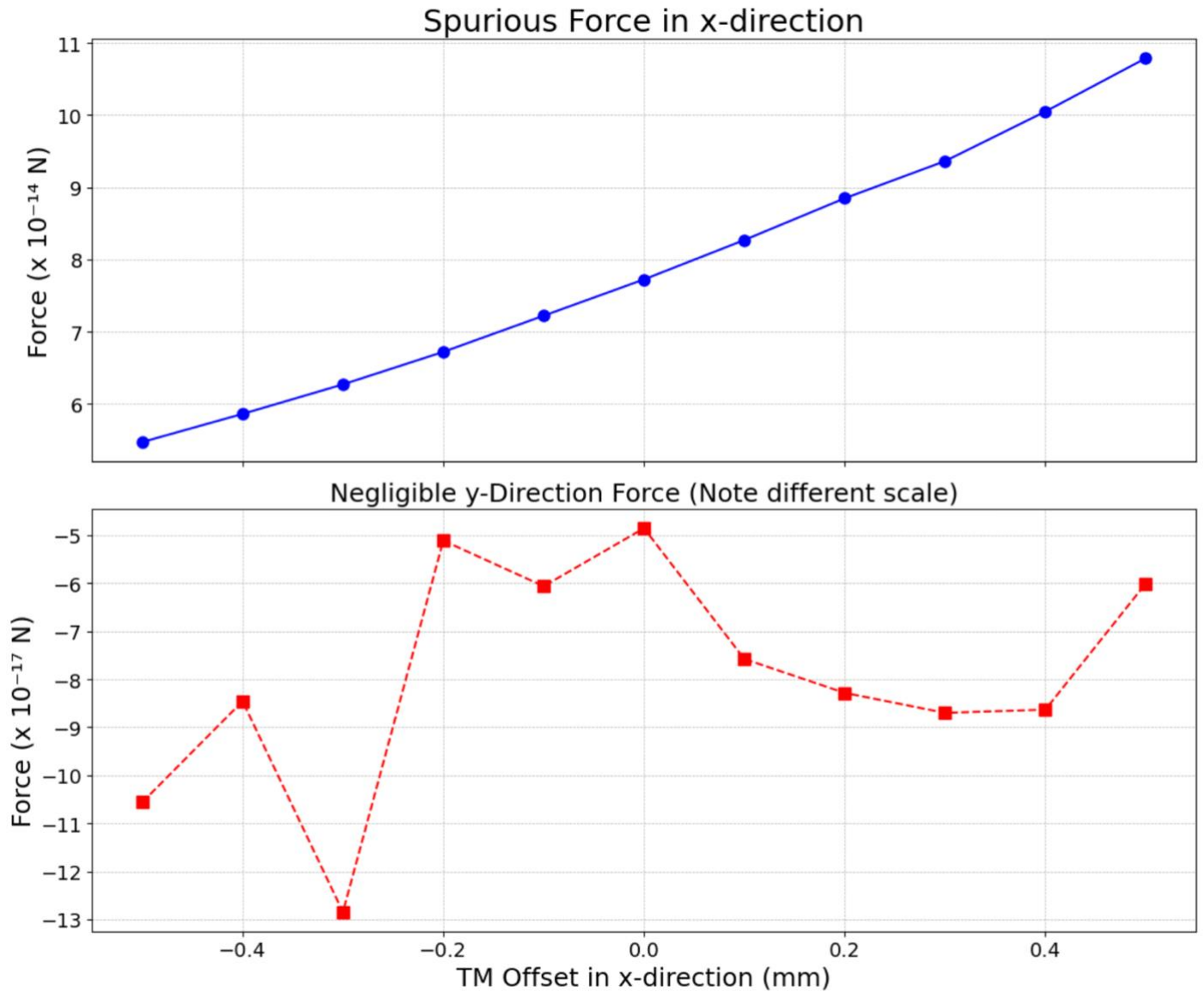


Figure 14: Simulated Force in x and y direction vs. TM x-offset

The data in [Figure 14](#) reveals two important findings. First, the force increases in a nearly linear fashion as the TM moves further from the center, demonstrating the direct coupling between geometric misalignment and spurious force. Second, and more critically the simulation calculates a non-zero force of $0.72 \times 10^{-13} \text{ N}$ even at a zero geometric offset (0 m). In a perfectly ideal system a centered TM would experience zero net force. This non-zero result demonstrates that the electrical asymmetry introduced by the millivolt-level patch potentials is sufficient on its own to generate a significant spurious force. The total force is therefore a combination of a baseline force from electrical asymmetries which is then modified by the growing geometric asymmetry as the TM is displaced. We also see very negligible fluctuating force in y-direction.

The static force of $\approx 10^{-13} \text{ N}$ has a direct consequence for the mission. This force acting on the 1.96 kg TM produces a DC acceleration $a = F/m$: $10^{-13}/1.96 = 5.1 \times 10^{-14} \text{ ms}^{-2}$. This static acceleration is more than ten times larger than the LISA mission's acceleration noise requirement of $3 \times 10^{-15} \text{ ms}^{-2} \text{ Hz}^{-2}$. This comparison demonstrates that even small realistic imperfections in geometry and surface potentials produce forces strong enough to overwhelm the gravitational wave

signal. This highlights why the active charge mitigation strategies discussed in Chapter 7 are essential for the mission's success.

8.2.2 Capacitance

Before relying on the force calculations, the model's fundamental electrostatic properties were validated. This was done by computing the total capacitance C_T of the test mass relative to its housing. This check is crucial because the spurious forces are directly proportional to the spatial gradient of capacitance $\partial C/\partial x$. An accurate capacitance model is a prerequisite for an accurate force model. A Stationary Source Sweep study was used to compute the system's mutual capacitance matrix, shown in [Table 3](#).

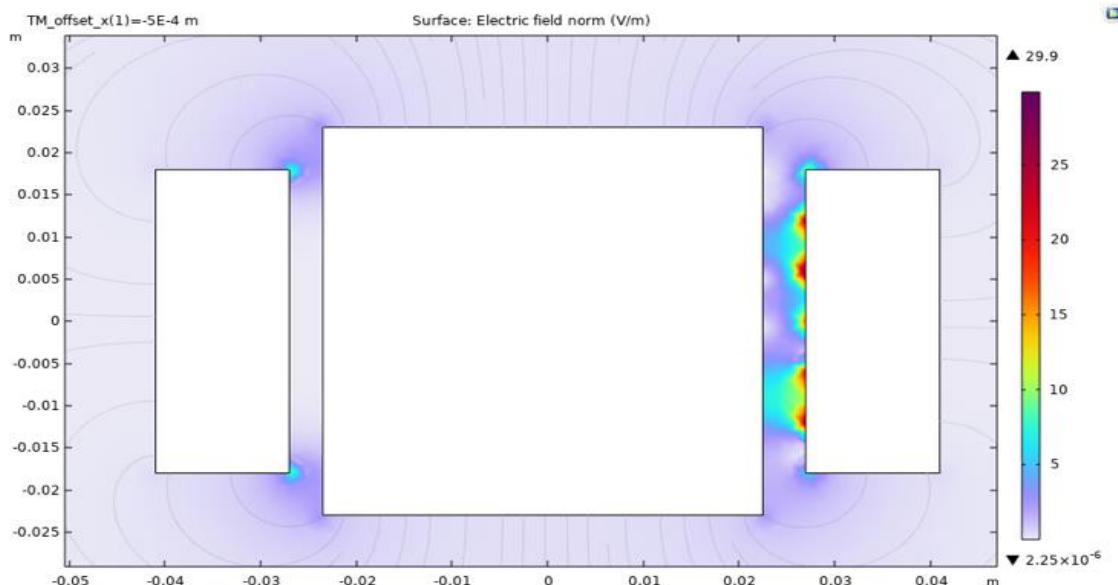
18.591	98.576	107.603
98.576	15.801	0.211
107.603	0.211	15.704

Table 3: Mutual Capacitance matrix (pF)

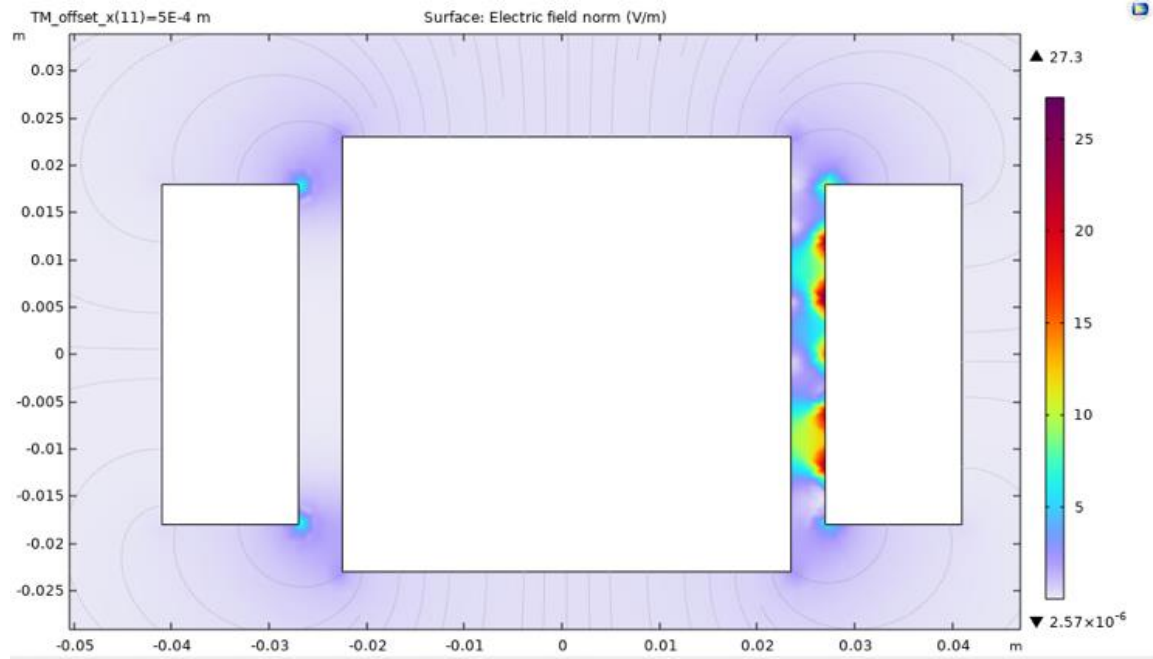
The capacitance of the TM relative to the other conductors is the first diagonal element of the matrix ($C_{m,11}$ in *COMSOL* terms). The simulated total capacitance is 18.59 pF. This value was compared to the known capacitance of the full 3D 18-electrode gravitational reference sensor which is approximately 34.2 pF [28]. The lower capacitance in the 2D model is an expected and logical result. This simplification only considers the electrodes along the sensitive x-axis and omits the capacitance contributions from the 16 other electrodes surrounding the TM in the full 3D assembly. The fact that the simulated value is the same order of magnitude as the real world sensor confirms that the model's representation of the core geometry particularly the 4 mm gap and 46 mm TM size is correct and scales properly. This successful validation of the model's basic electrical properties gives high confidence that its subsequent force and field calculations are physically meaningful.

8.2.3 Electric Field, Electric Potential and Surface Charge density

The force (Section 8.2.1) and capacitance (Section 8.2.2) results mutually validate the COMSOL model. The realistic capacitance calculation confirms the model's electrostatic properties by verifying its primary finding; femto Newton level spurious forces are generated by realistic misalignments and patch potentials. This confirms the 2D COMSOL approach as a valid tool for quantifying force noise. Therefore beyond numerical quantification the model's graphical outputs provide a clear visualization of the physical origin of this force.



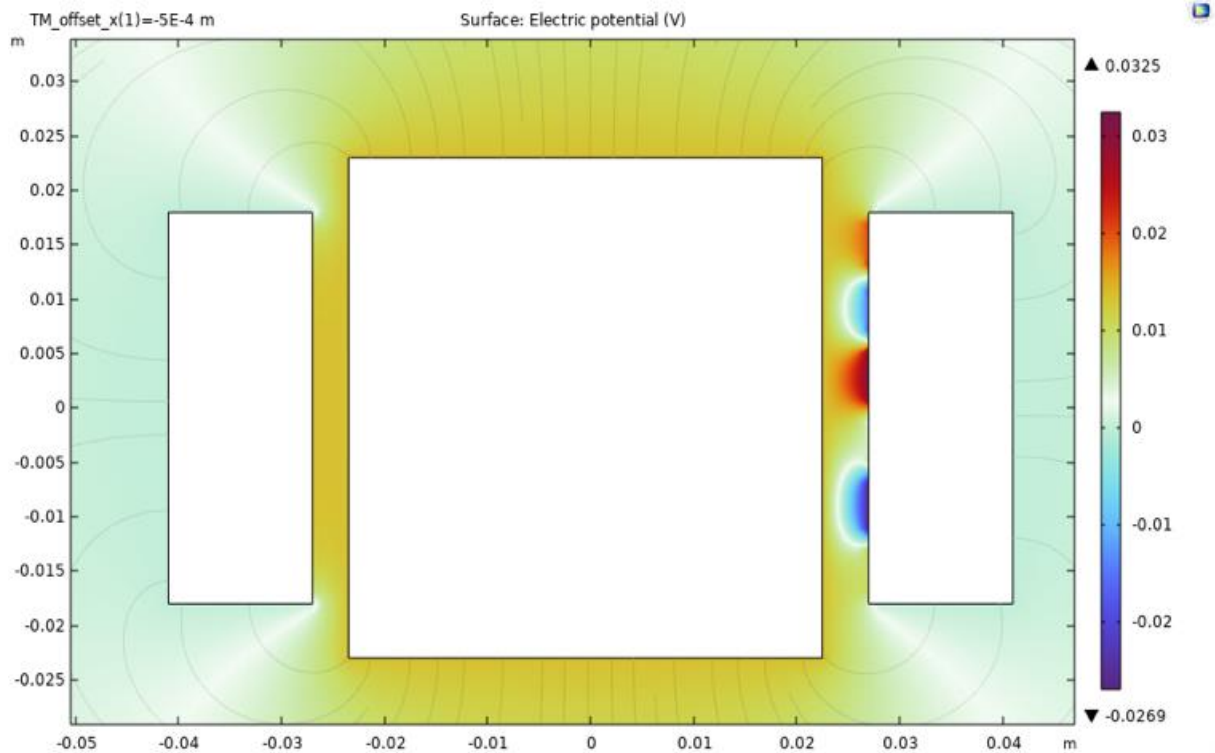
i)



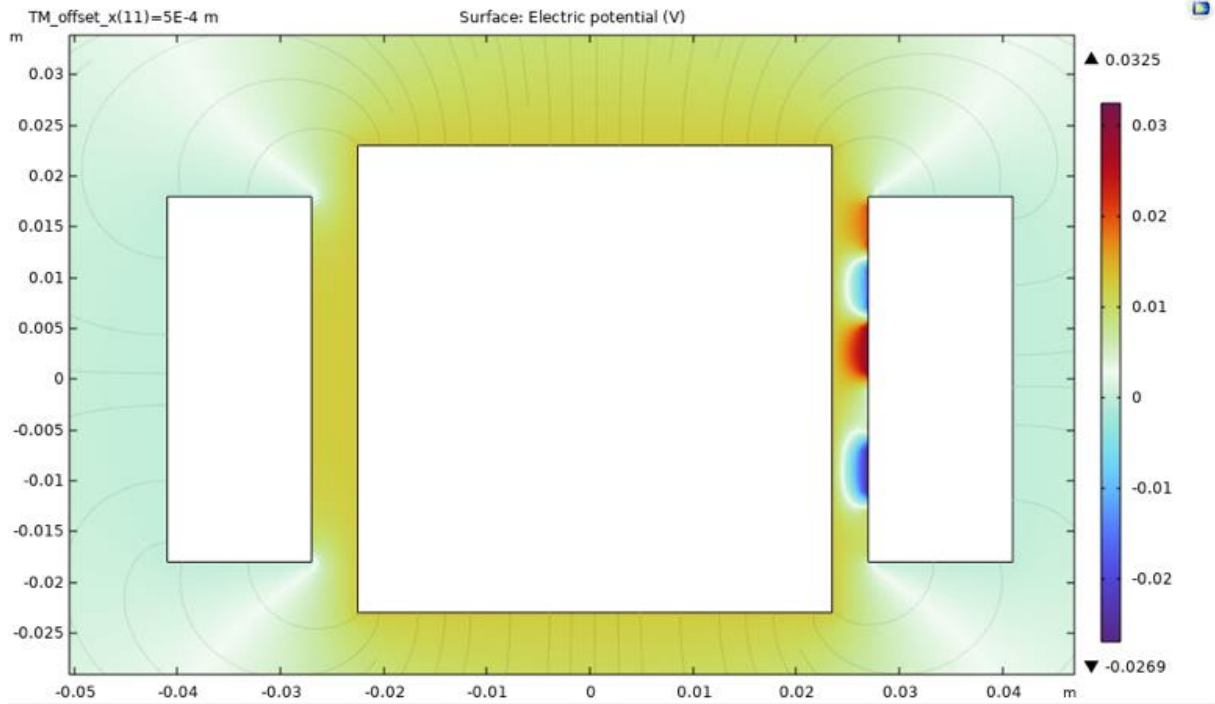
ii)

Figure 15: The electric field norm (V/m) and field lines for a TM offset of i) -0.5 mm ii) 0.5 mm

The electric field plot shown in [Figure 15](#) reveals the field imbalance in the electrostatic environment. The field streamlines are visibly denser in the narrower gap between the TM and the right electrode. This imbalance is the direct cause of the net electrostatic force pushing the TM (in this case to the left). This asymmetry is a combined result of the geometric offset and the non-uniform potentials on the right electrode surface. The electric field plot is a bit more intense (noted from maximum of scale on the right) when TM is drifting towards right electrode.



i)



ii)

Figure 16: Electric potential (V) of the GRS model for offsets i) -0.5 mm ii) 0.5 mm. The TM (center) is at a positive floating potential due to its net $10^7 e$ charge while the patch potentials are visible on the right electrode.

The electric potential plot in [Figure 16](#) confirms the initial conditions of the simulation. This plot shows the TM held at a positive floating potential (yellow/red), consistent with its net positive charge of $10^7 e$. The left electrode is at 0 V (blue), and the individual patch potentials are visible as distinct red and blue segments on the surface of the right electrode. Difference in potential plots at both offsets is negligible.

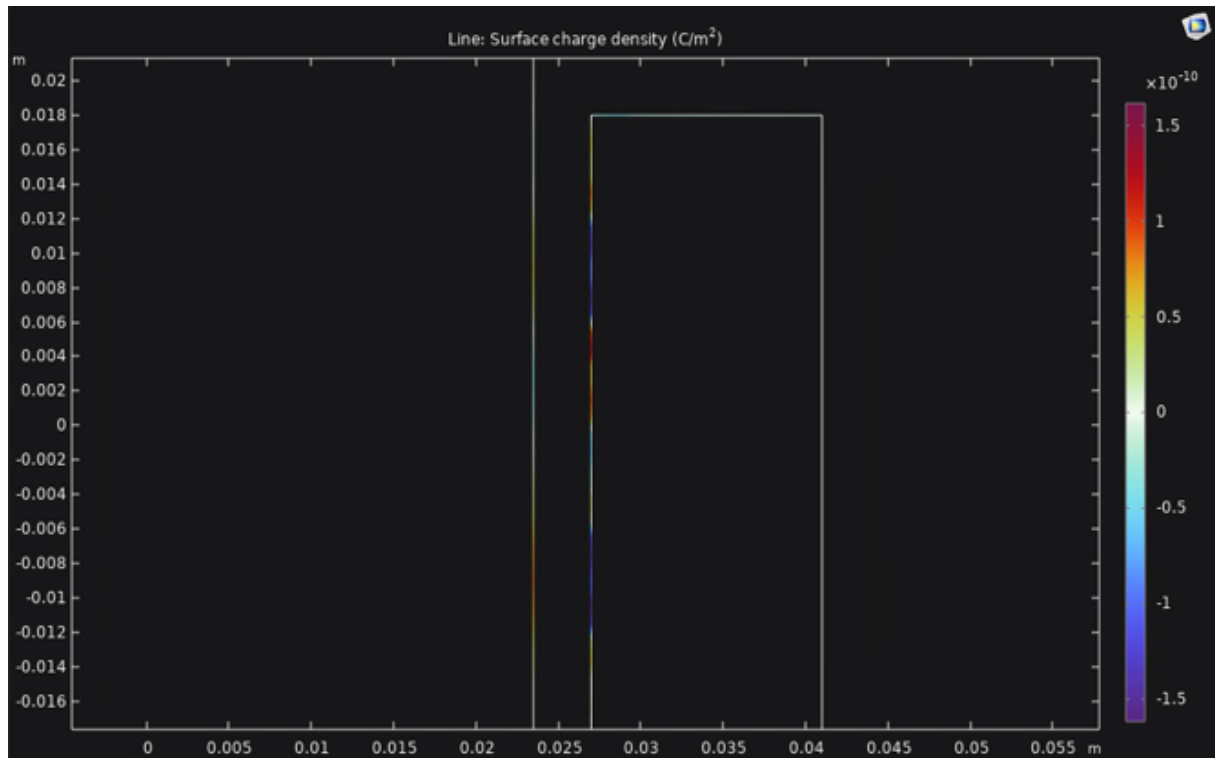


Figure 17: Surface charge density C/m^2 on the conductor surfaces. The plot shows how the patch potentials on the right electrode induce a non-uniform charge which in turn induces an opposite charge on the TM.

Finally the surface charge density plot in [Figure 17](#) reveals the microscopic source of the force. This plot shows the charge density on the surfaces of the conductors. The millivolt patch potentials applied to the right electrode (which are too small to see on the main potential plot) cause a non-uniform charge distribution on that electrode's surface. This in turn induces an opposite charge separation on the facing surface of the test mass. This induced charge pattern is the microscopic source of the asymmetric electric field that exerts the net force on the charged TM.

Together these visualizations provide a clear understanding of the problem. They break down the abstract force equation into an illustration of three distinct and unavoidable factors:

- A net TM charge q_T which sets the TM's potential.
- A geometric misalignment δd which creates an asymmetric field.
- Surface potential variations Δ_x which create an additional electrical asymmetry.

These factors work in combination to produce the final unwanted spurious force that the mission must mitigate.

9 Further research outlook

Future work should focus on enhancing the accuracy and efficiency of charge mitigation systems. One possible area for development is the use of advanced simulation software such as COMSOL Multiphysics to create complex and real models of the test mass and its electromagnetic environment as development of fully featured modeling and simulation tools is essential to accurately predict charging processes. These simulations will allow for the testing and optimization of novel charge control strategies such as improved UV light discharge patterns or improved shielding geometries under a wide range of simulated space weather conditions. This approach can refine the technologies required for the LISA mission ensuring robust performance against unpredictable charging events.

10 Conclusion

This review confirms that the electrostatic charging of test masses is a critical challenge that directly impacts the sensitivity of space based gravitational wave detectors. The primary sources of charging have been identified as galactic and solar high energy particles. Through a combination of passive mitigation such as structural shielding and careful material selection and active charge control using UV light the LISA PF mission successfully managed the TM potential to required levels. This achievement demonstrates that the problem of TM charging is well understood yet could be still improved when aiming for higher sensitivity, providing crucial confidence in the technological readiness for the forthcoming LISA mission. It was also shown that using simulation tools like COMSOL you can model the required environments accurately. This would allow us to create further complex models for future LISA-like missions that can be help understand the charging process and the mitigation systems together in the space environment.

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